



Outline

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Outline

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2	Role of LEF in physical design
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What is a LEF File?

LEF (Library Exchange Format) is an industry-standard format that describes the abstract physical representation of standard cells and IP blocks

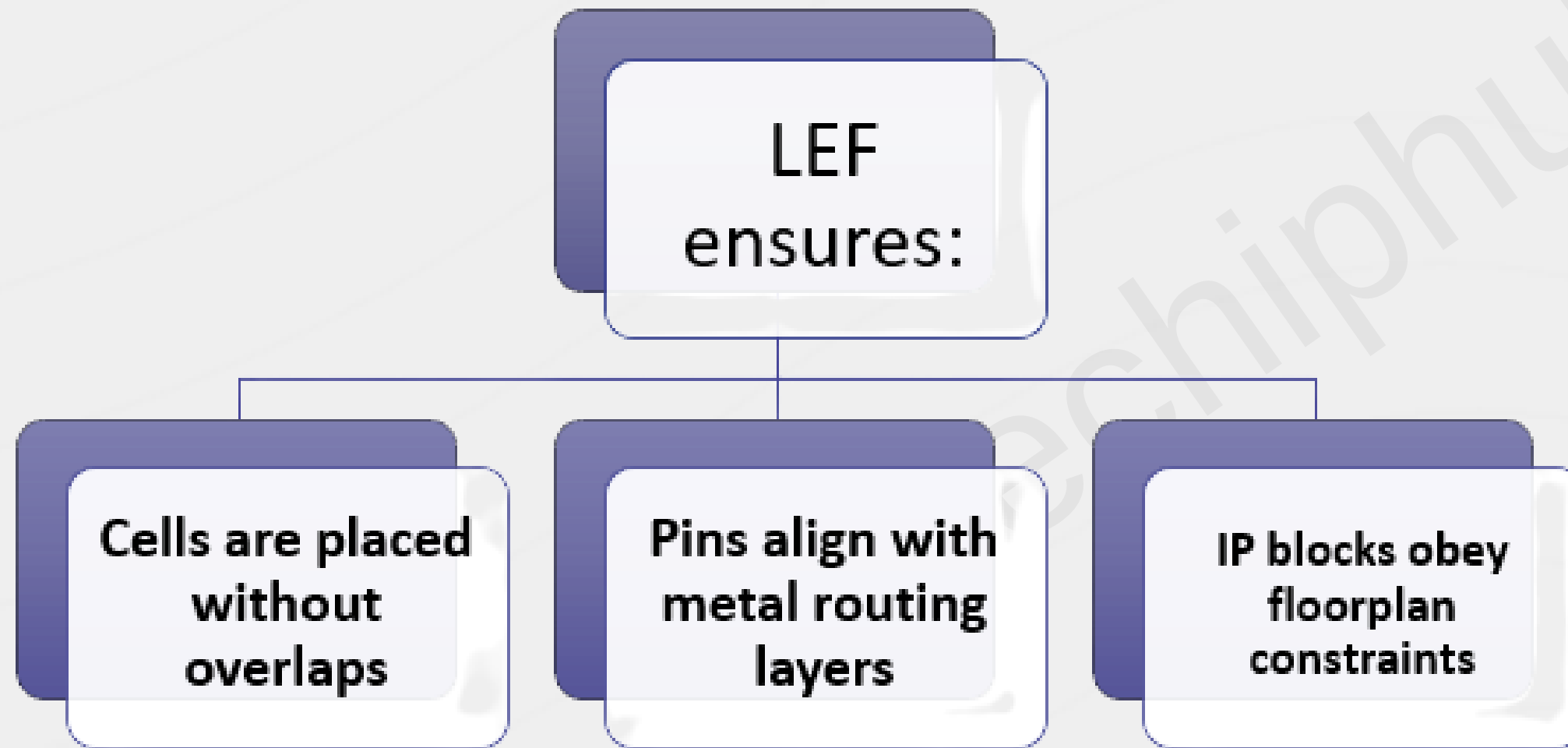
- It provides:
 - ▪ Cell boundary dimensions
 - ▪ Pin geometry and access points
 - ▪ Routing layer usage
 - ▪ Placement grid and alignment rules
 - LEF intentionally avoids detailed polygons to enable fast placement and routing.

➔ Role of the LEF in Physical design?

After synthesis, tools understand logical intent, but not physical feasibility.

- Determine legal placement locations
- Align pins with routing layers
- Enforce row and site compatibility
- Prevent overlaps and routing conflicts

➔ Role of the LEF in Physical design?



Without LEF, physical implementation is not possible.

LEF vs. LIB – What's the Difference?

LIB (.lib): Electrical View

- Timing arcs and delays
- Power characterization
- Logical functionality

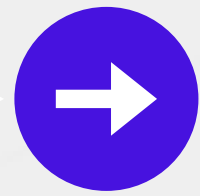
➔ LEF vs. LIB – What's the Difference?

LIB (.lib): Electrical View

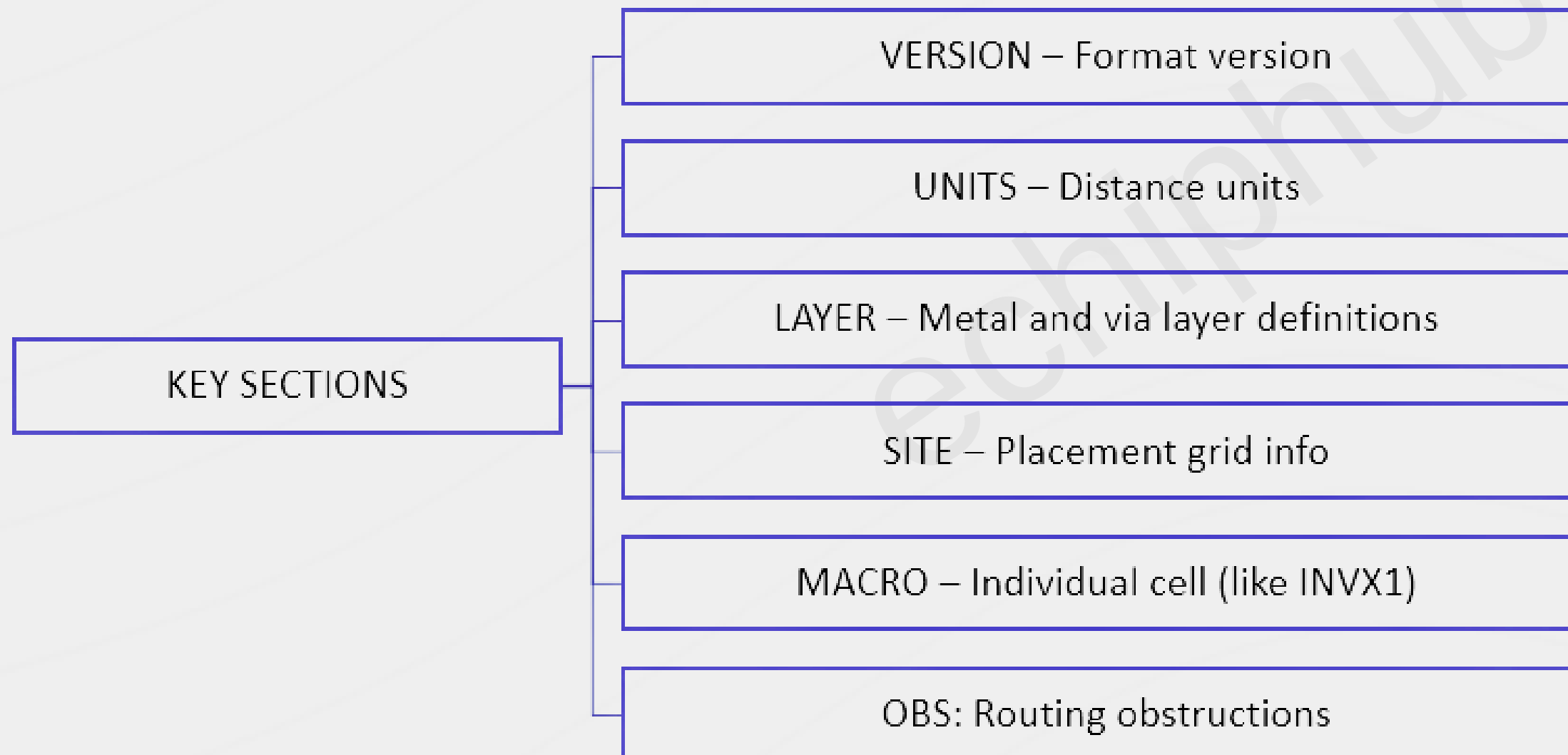
- Geometry and pin placement
- Metal layers and routing access
- Placement grid and symmetry

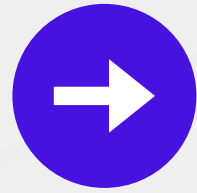
Both files describe the same cell from orthogonal abstraction domains

**Both files describe the same cell
but from different perspectives**



Key LEF sections: MACROS, SITE, LAYER





Example of LEF snippet INVX1

```
MACRO INVX1
  CLASS CORE ;
  ORIGIN 0 0 ;
  SIZE 1.2 BY 2.0 ;
  PIN A
    DIRECTION INPUT ;
    PORT
      LAYER M1 ;
      RECT 0.1 0.4 0.2 0.6 ;
    END
  END A
  PIN Y
    DIRECTION OUTPUT ;
    PORT
      LAYER M1 ;
      RECT 0.9 1.0 1.0 1.2 ;
    END
  END Y
END INVX1
```

➔ Understanding MACRO Parameters

CLASS	'CORE' for logic gates, 'PAD' for IO cells
SIZE	Width × Height of cell (in microns or units)
ORIGIN	Bottom-left reference corner
PIN	Input or output terminals
LAYER	Which metal layer the pin connects to
RECT	Physical bounding box of the pin

➔ LEF in floorplanning and placement

SITE defines the cell row height and
grid structure

```
SITE coreSite  
  SIZE 1.0 BY 2.0 ;  
  CLASS CORE ;  
  SYMMETRY Y ;  
END coreSite
```

- Every standard cell is snapped to a SITE grid for alignment.
- Ensures uniform placement during floorplanning.

LEF helps physical design tools:

- Place cells along legal rows
- Align pins for proper routing
- Avoid overlap and routing blockages
- Define I/O block locations

LEF vs. DEF – Key Differences

LEF

Library abstraction

Reusable across designs

Describes cell geometry

Technology view

DEF

Design instantiation

Design-specific

Describes cell placement

Implementation view

Both are needed for layout, placement, and routing.

➔ LEF in Open-Source Tools

Tools that use LEF:

OpenLane	OpenRoad inside openlane	KLayout (for visual LEF/GDSII mapping)	Magic VLSI (LEF-based layout generation)
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Students can use LEF files from Google
SkyWater130 PDK.

➔ Summary and Best Practices

SUMMARY

LEF gives abstract physical views of cells

Used during placement and routing

Defines geometry, pin location, routing layers

Complements .lib for complete cell definition

BEST PRACTICES

Ensure LEF matches .lib in functionality

Use correct layer/pin sizes for DRC-clean designs

Learn to read MACRO and SITE sections carefully



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SoC Teamup

Thank you !

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